6. MIDDLE EAR DISORDER

PATHOGENESIS OF MIDDLE EAR DISEASES

00:00:55

- All middle ear diseases occur due to eustachian tube obstruction
- Causes of Eustachian Tube Obstruction
  - Nasal Causes
    → DNS
    → Nasal polyps
    → Allergic rhinitis
    → Nasal packing for epistaxis
    → Nasal foreign body
  - Nasopharyngeal Cause
    → Any pathology affecting **nasopharynx ventilation** can obstruct the eustachian tube.
  - Palatal Causes
    → Cleft palate
    - Eustachian tube opening depends on Tensor veli palatini and Levator veli palatini muscles
    - If these muscles are weak → tube cannot open → middle ear ventilation decreases.

Nasal / Nasopharyngeal / Palatal pathology
↓
Eustachian tube obstruction
↓
Negative pressure in middle ear
↓
Middle ear effusion

- Infected effusion : Suppurative otitis media
- Non Infected effusion : Non suppurative otitis media
- < 4 weeks: Acute Suppurative Otitis Media (ASOM)
- > 12 weeks: Chronic Suppurative Otitis Media (CSOM)

ACUTE SUPPURATIVE OTITIS MEDIA (ASOM)

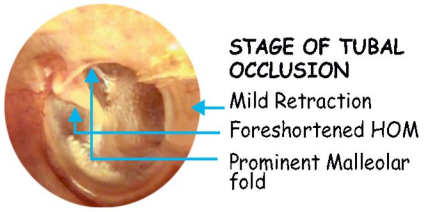
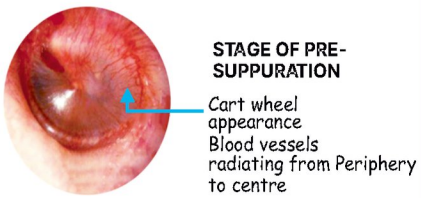
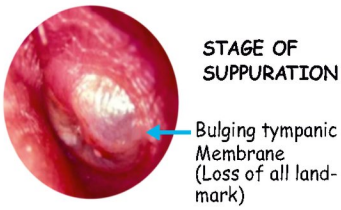
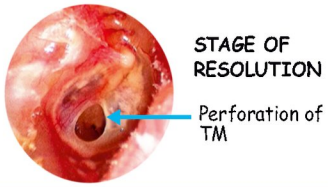
AIIMS 2020
NEET PG 2018, 2022
INICET 2022, 2024, 2025
FMGE 2018, 2022, 2024, 2025

- Predisposing factors leading to eustachian tube blockage.
  - Recurrent attacks of common cold,URTI
  - Exanthematous fever like measles,diphtheria,whooping cough
  - Chronic rhinitis and sinusitis
  - Allergy
  - Nasopharyngeal tumors
  - Nasal packing for epistaxis
  - Cleft palate

YOURWISH

YOURWISH

STAGES OF ASOM

Stage	Pathology	Symptoms	Signs
1.TUBAL OCCLUSION	Tubal oedema of ET leading to negative pressure and retraction of TM	Ear blockMild deafness	Mild retraction of tympanic membrane 
2.PRESUPPURATION	Invasion of organisms and mild inflammation	Severe ear pain Hearing loss Fever	Blood vessels radiating from periphery to centre Cartwheel appearance 
3.SUPPURATION	Frank pus formation	Excruciating ear pain High-grade fever Hearing loss	Bulging tympanic membrane Pus discharges in pulsatile form : Pulsatile otorrhea This pus reflects the light: light house sign 
4.RESOLUTION	TM rupture with release of pus	Pus drains out Tympanic membrane perforation in anteroinferior quadrant	Pus drains out TM perforation 
5.COMPLICATIONS			

- MC organism : Streptococcus pneumoniae
- Symptoms : Child with history of URTI, Ear pain, Fever, Hearing loss
- Signs: TM retraction, Cartwheel appearance, Bulging drum, Pulsatile otorrhea, Lighthouse sign
- Investigations : Usually not required
- Diagnosis : History + Clinical examination
- Rx :

- Amoxicillin/Ampicillin (8–10 days)
- Supportive : Nasal/oral decongestants, Analgesics, Antipyretics
- Myringotomy
  → Indications
  - Excruciating pain + bulging drum
  - Incomplete resolution despite antibiotics
  - Persistent effusion >12 weeks
  - ASOM with facial palsy

  → Curvilinear incision in posterior inferior quadrant

Patient presents with c/o acute onset ear pain Wo URTI

Next step

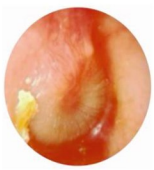
On Examination
TM cartwheel, red congested, Pulsatile otorrhea, Light house sign.

Cartwheel TM

Diagnosis
ASOM
MC route-Eustachian tube

Next step

Management
Antibiotics (Pneumococcus MC organismi), myringotomy (Poste anterior quad, done if TM fill, bulging or facial palsy



Posteroinferior quad, curvilinear Myringotomy

NON-SUPPURATIVE OTITIS MEDIA (SOM)

00:11:33

- Definition: Accumulation of non inflammatory exudate in the middle ear cavity
- Common in 2-6 years of age
- Occurs due to:
  - Nasal pathology
  - Nasopharyngeal pathology
  - Palatal pathology

Condition	Age Group	Cause
Bilateral SOM	Children	Adenoid hypertrophy
Unilateral SOM	Juveniles	Juvenile Nasopharyngeal Angiofibroma (JNA)
Unilateral SOM	Older Age	Nasopharyngeal carcinoma (NPC)

ETIOLOGY

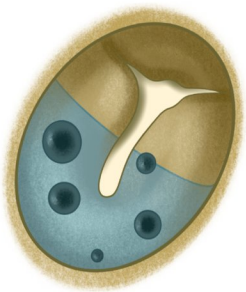
- Tubal occlusion due to various nasopharyngeal causes like adenoids benign and malignant tumour
- Allergy : Mucosal edema of eustachian tubal orifice
- Hypogammaglobulinemia predisposes to inflammatory response in the middle ear cavity

PATHOLOGY

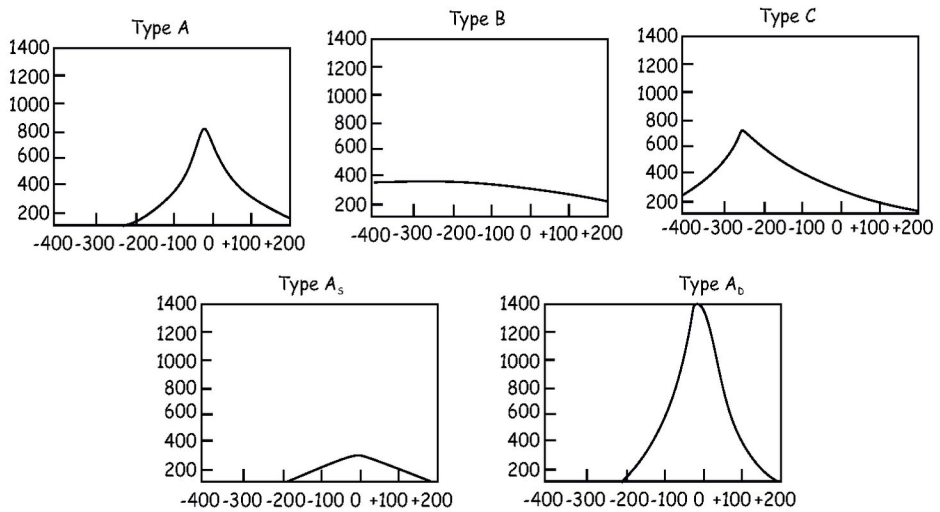
- Sterile effusion in middle ear
- Fluid is non-infective
- No pus formation

CLINICAL FEATURES

- Symptoms
  - Painless
  - Block sensation of the ear
  - CHL
  - Delayed speech
  - Poor school performance
  - Autophony
  - Nasal discharge and obstruction
  - There is no pain and no fever.
- Signs
  - Dull lusterless tympanic membrane with absent light reflex
  - Retracted tympanic membrane
  - Presence of fluid level
  - Air bubbles
  - Decreased mobility of the tympanic membrane



Hearing Tests	Findings
Tuning Fork Test	Rinne negative Weber lateralized to affected ear Indicates: Conductive hearing loss
Pure Tone Audiometry	Conductive hearing loss
Tympanometry(IOC)	Type B Tympanogram (indicates middle ear effusion)



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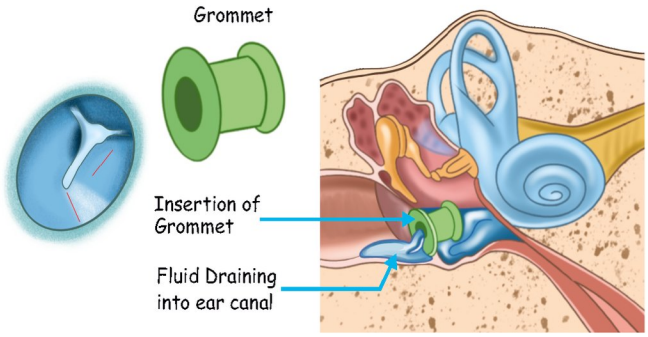
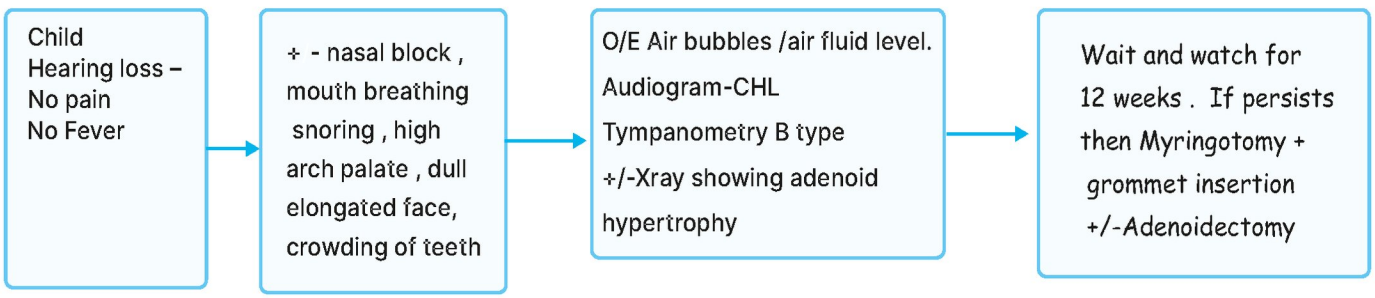
TREATMENT

Situation	Management
Hearing loss only	• Wait & watch for 3 months
Persistent effusion (If no change even after 3 month)	• Myringotomy +/- grommet insertion ○ Radial incision in the anteroinferior quadrant 
Adenoid hypertrophy present (Mouth breathing, Snoring, High arched palate, Failure to thrive)	Myringotomy + grommet + adenoidectomy Fees:

- Prevention of secretory otitis media :
  - Anti allergy treatment
  - Decongestants
  - Middle ear aeration repeated valsalva or politzerization



CHOLESTEATOMA

00:17:40

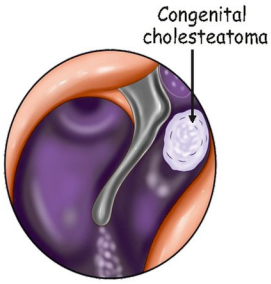
- Definition : Presence of stratified squamous epithelium in middle ear.

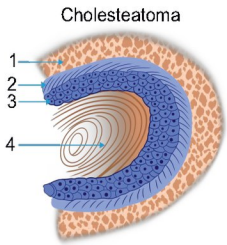
AIIMS 2020, FMGE 2021, 2025
INICET 2021, NEET PG 2019, 2020

Feature	Congenital Cholesteatoma	Acquired Cholesteatoma
Pathogenesis	EAC epithelium trapped in middle ear during development (cleft in pouch)	• Primary: No prior pathology • Secondary: TM perforation or retraction pocket
Common Sites	Middle ear, Petrous apex, Cerebellopontine angle	Attic, Posterosuperior pars tensa

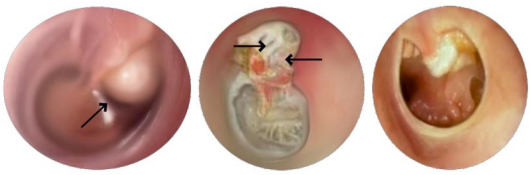
YOURWISH

YOURWISH

Otoscopy	Whitish mass behind intact TM TM normal 	TM perforation or retraction pocket with debris
Levenson's Criteria	White mass with intact TM, no prior otorrhea/surgery/disease,	Not applicable
Laterality	Often unilateral (commonly left side in some series)	Commonly right side



- Cholesteatoma is unsafe as it spreads like tumor because of bone destroying property
- Collagenase, alkaline phosphatase, proteolytic enzymes cause bone erosion
- Spreads along least resistance path
- MC involves lateral semicircular canal
- Symptoms
  - Hearing loss (most common)
  - Scanty foul-smelling discharge
- Diagnosis
  - Otoscopy under microscope: Whitish keratinous plaque (hallmark)
  - IOC: CT scan temporal bone (assess disease extent, bone erosion; no contrast)
  - Audiological tests : CHL
- Rx:
  - Medical therapy does not cure
  - Surgery of choice is Modified Radical Mastoidectomy (MRM)



Cholesteatoma Hearer

- Mechanism: Cholesteatoma first erodes incus (usually long process)
- Effect: Cholesteatoma bridges ossicular gap, allowing sound conduction
- Clinical significance:
  - Before surgery: Hearing relatively preserved
  - After cholesteatoma removal: Gap appears, postoperative hearing worsens

Round Window Shielding Effect

- Normal sound transmission: TM → Malleus → Incus → Stapes → Oval window → Scala vestibuli → Scala tympani → Round window
- Sequential vibration: Oval window first, round window later
- TM perforation effect: Sound reaches both windows simultaneously → Vibrations cancel each other→ Hearing loss

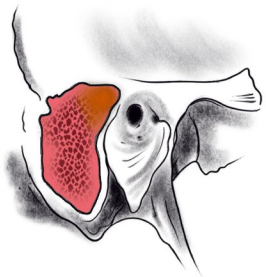
	Discharge effect on Hearing
Dry ear	Worse hearing(sound hits both windows simultaneously)
Ear with discharge	Better hearing (discharge shields round window)

Feature	Details
MC Site of Origin	Prussak's space(Between pars flaccida and neck of malleus)
MC Site of Recurrence	Sinus tympani(Aka Infrapyramidal space)
MC Ossicle Eroded	Long process of incus(Lenticular process)
MCCanal Involved	Lateral semicircular canal

Mastoidectomy

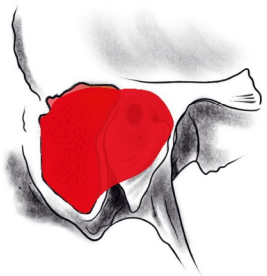
- Types
  - Cortical Mastoidectomy Schwartz operation
  - Modified Radical Mastoidectomy

CORTICAL MASTOIDECTOMY



- Procedure
  - Mastoid air cells removed
  - Mastoid converted into single cavity
  - Posterior canal wall kept intact
- Aka Intact canal wall , Schwartz operation

MODIFIED RADICAL MASTOIDECTOMY (MRM)



- Procedure
  - Mastoid air cells removed
  - Posterior canal wall removed
  - Mastoid + middle ear + external auditory canal connected into single cavity
- AKa: Canal wall down surgery
- After MRM, meatoplasty is done to reshape the external auditory canal

CHRONIC SUPPURATIVE OTITIS MEDIA (CSOM)

00:30:11

- Definition: Chronic middle ear infection of whole or part of cleft with ear discharge and permanent perforation
- Characteristic Feature: Permanent tympanic membrane perforation(Perforation edges lined by squamous epithelium preventing healing)

Types OF CSOM

Old Classification	New Classification
Tubotympanic	Mucosal type
Atticoantral	Squamosal type

MUCOSAL CSOM

- Inactive mucosal - dry,no pus
- Active mucosal - pus present

SQUAMOSAL TYPE (ATTICOANTRAL)

- Inactive - dry with cholesteatoma associated
- Active- pus present

Feature	Mucosal Type (Tubotympanic/Safe)	Squamosal Type (Atticoantral/Unsafe)
Old/New Classification	Tubotympanic / Mucosal	Atticoantral / Squamosal
Age	Children	Adult
Site of Perforation	Pars tensa (central perforation)	Pars flaccida / Posterosuperior pars tensa
Etiology	Eustachian tube dysfunction, Nasopharyngeal pathology	Cholesteatoma formation
Clinical Features	Profuse mucoid discharge Non-foul smelling Non-blood stained	Scanty foul-smelling discharge Blood stained
Disease Area	Anterior inferior middle ear	Posterior superior middle ear
Subtypes	Inactive: Dry perforation Active: Discharging ear	Inactive: Cholesteatoma without discharge Active: Discharging ear
Discharge	Profuse	scanty
Discharge smell	Odourless and mucoid	Foul and purulent
Safety	Safe	Unsafe
Bone Erosion	Rare	Common
Cholesteatoma	Absent	Present
Osicular necrosis	Rare	Common (Most commonly long process of incus)
Hearing loss	Mild moderate	Moderate severe
Tegmen tympani erosion	Rare	Common
Intracranial complication	Rare	Common
Polyp	Pale	Red,fleshy
Tympanoscelrosis	Common	Rare
Cholesterol granuloma	Rare	Common

YOURWISH

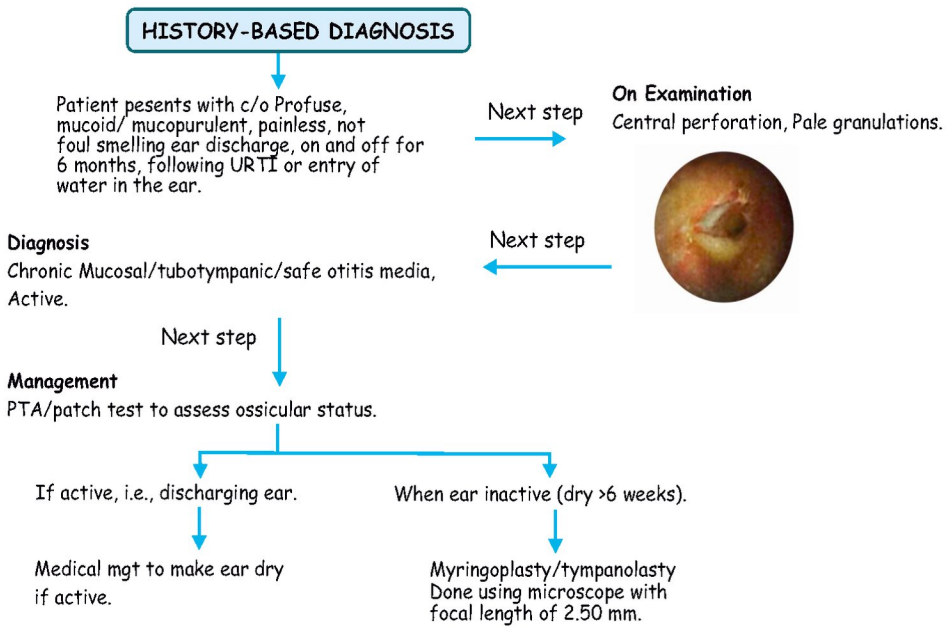
Treatment	Treatment of primary pathology is the main stay + Reconstruction surgery (Myringoplasty / Tympanoplasty)	Treatment is ear surgery depending on the extent of disease + Reconstruction surgery (Myringoplasty / Tympanoplasty/ reconstruction of ossicular chain)

INVESTIGATIONS

Investigation	Purpose
Otoscopy	Detect perforation
Audiometry	Assess degree of hearing loss
Culture and sensitivity	Ear discharge
Mastoid xray	Sclerotic and extent of bone destruction
CT temporal bone	Disease extent

MEDICAL TREATMENT

- Aural toilet : Debris and aural polyp should be removed before local treatment . Do not avulse in aural polyp
- Ear drop : neomycin + polymyxin + chloromycetin or gentamycin is used
- Systemic antibiotics : Limited use only in acute exacerbation
- Keep ear dry



TUBERCULAR OTITIS MEDIA

00:38:00

- Painless foul smelling discharge
- Multiple tympanic membrane perforations
- Facial nerve paralysis possible
- Severe hearing loss out of proportion of symptoms

Q. Which of the following signs is not associated with otitis media with effusion? INICET May 2024

A. Mostly bilateral

B. AD curve on tympanometry with conductive hearing loss

C. Adenoid hypertrophy is cause of SOM

D. Mostly resolves on its own

Answer: **B**

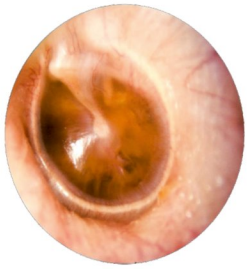
Q. You see a child with delayed speech development as part of a health visit at a local school. The otoscopic appearance of the ear is shown. Conductive hearing loss of 20 dB is detected in audiometric tests. What is your diagnosis? FMGE June 2022

A. Acute suppurative otitis media

B. Serous otitis media

C. Chronic suppurative otitis media

D. Glomus tumor

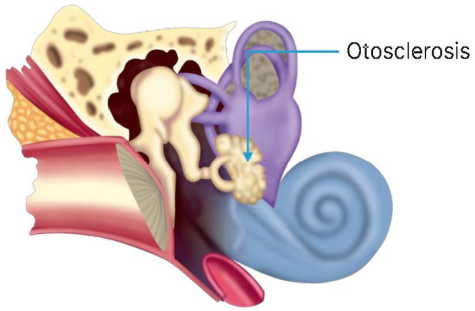


Answer: **B**

7. OTOSCLEROSIS

OTOSCLEROSIS

00:00:44

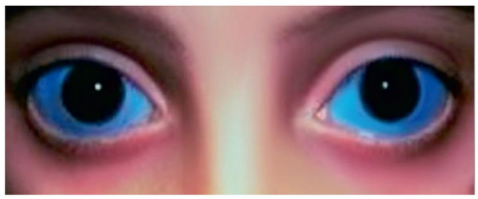


- Excessive bony deposition around the,
  - Foot plate of stapes: Stapedial type
  - Round window: Cochlear type
- Stapedial:
  - M/c Site: fistula ante fenestrum (anterior to the foot plate of stapes)
- Cochlear:
  - M/c Site: starts at the round window and goes to the scala tympani

STAPEDIAL TYPE:

Aetiology:

FMGE 2022



- Hereditary: autosomal dominant with incomplete penetrance
- Sex: F>M, but in India it is male-dominant.
- Age of onset: 20-40 years
- Increases in pregnancy and during menopause
- Measles is a provocative factor
- Vanderhoove syndrome: otosclerosis + blue sclera + osteogenesis imperfecta

SYMPTOMS OF OTOSCLEROSIS

Stapedial

- M/c presenting feature: conductive hearing loss - bilateral
- Paracusis Willisii: A condition where the patient hears better in a noisy environment as compared to a silent environment.

Cochlear:

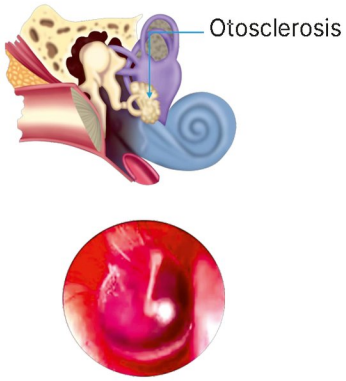
- Sensorineural hearing loss
- Tinnitus

	STAPEDIAL	COCHLEAR
Incidence	Common	Less common
Hearing loss	Conductive	Sensorineural
Tinnitus, Vertigo	Rare	Common
Paracusis willisii	Common	No

Signs

- The tympanic membrane is normal (90%)

- The tympanic membrane is mobile
- Schwartz sign (10%): reddish/pinkish hue seen
  - Active lesions have a flamingo-pink hue on the promontory and can be seen through an intact tympanic membrane
- **NOTE**: Classic presentation is gradually progressive conductive hearing loss, which is B/L in a middle-aged female.



Tests

Stapedial Otosclerosis

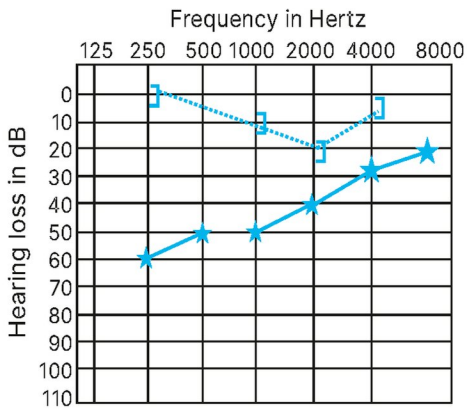
AIIMS 2018

Rinne	Negative
Weber	Lateralized to abnormal ear
Absolute bone conduction	Normal
Gelles test	Negative
Tympanometry (IOC)	As-type graph
Stapedial reflex	Absent
Pure tone audiometry	Carhart's notch (dip in BC curve at 2000 Hz)

AUDIOMETRY

FMGE 2024
INICET 2019, 2024

- Ac: lower frequencies are affected
- Bc: maximum dip is seen at 2000 Hz which is called Carhart's notch
- Reverses after stapedectomy



Cochlear

- Cookie Bite Curve: A dip in the midfrequency range of both AC and BC is called the cookie bite curve

Treatment

Medical

- Hearing aid
- Sodium fluoride: It is given to arrest focus and prevent further sensorineural loss due to cochlear otosclerosis
  - MOA:
    → Inhibits the osteoclastic and osteoblastic activities and inhibits the release of proteolytic enzymes
    → Only prevents further progression.
  - Indication: only in active focus
  - Contraindication:

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→ Pregnancy and lactation
→ Renal disease
→ Rheumatoid arthritis
→ Skeletal fluorosis

Surgical:

- Stapedectomy with placement of prosthesis
- Stapedotomy with placement of prosthesis
- Lempert's fenestration operation: reserved only for cases where the footplate cannot be mobilized during a stapedectomy.

Indication for surgery	Contraindications for surgery
The hearing threshold is 30 dB, or A-B gap is 25 db or more	Only hearing ear
If there is bilateral otosclerosis, then worst ear is operated first	Pregnancy
	Occupational: pilots and divers

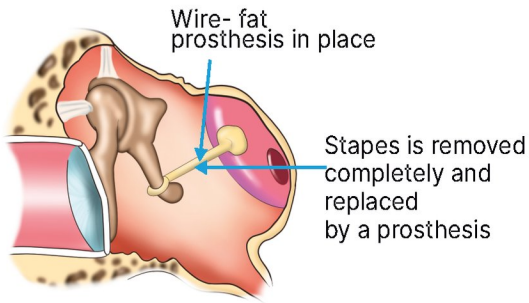
LOCAL ANAESTHESIA VS GENERAL ANAESTHESIA

- Local is preferred for stapes surgery, as intraoperative assessment is possible.
  - Vertigo could be assessed.
  - Facial nerve functions can be assessed

STAPEDECTOMY/STAPEDOTOMY

- Stapedectomy: Stapes is removed completely and replaced by a prosthesis
- Stapedotomy: A hole is made in the foot plate of the stapes, and from the incus to the oval window, an opening is made
  - Possibility for a tear in the oval window → leakage of perilymph → a fat plug is put in to prevent further leakage.
  - Treatment of choice

00:13:28



MANAGEMENT

Mature

- Stapedotomy (Local Anesthesia preferably) + Teflon piston (worst ear 1st) after cutting the stapedius tendon and removing stapessuperstructure the piston is put between incus long process and opening on stapes footplate.
- Other option Hearing aid, cochlear implant.

Active

- NaF (Sodium Flouride)→ increases maturation of focus by inhibiting osteoclastic and proteolytic activity and acceleratingosteoblastic activity

History-Based Diagnosis

- 20-30 years presents with b/l progressive hearing loss, which she noticed first after her pregnancy, with a similar history in her mother. Hearing better in noise (paracusis wilisii)

YOURWISH

On Examination:

- TM: normal pearly gray (90%)
- Active cases: flamingo pink (10%), a/k/a the Schwartz sign

Diagnostic Tests:

- TFT: CHL, gelle negative
- PTA: CHL, Cahart's Notch (BC dip at 2000 Hz)
- IOC: tympanometry - As
- Stapedial reflex: absent

PYQs

00:16:59

Q. A 45-year-old woman presents with progressive hearing loss that has been worsening over the past few years. Audiometry confirms conductive hearing loss, and a diagnosis of otosclerosis is made. What is the surgical treatment of choice for her condition?

FMGE Jan 2024

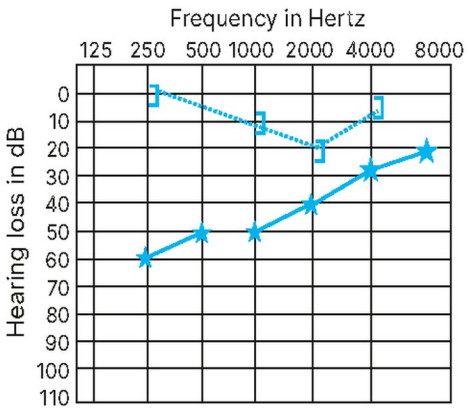
A. Stapedectomy with the placement of a prosthesis
B. Stapedotomy
C. Stapes mobilization
D. Lempert's fenestration operation

Answer: **A**

Q. An audiogram of a 30-year-old female is given. What is the diagnosis?

FMGE June 2024

A. Otosclerosis
B. Menieres disease
C. Noise-induced hearing loss
D. Presbycusis



Answer: **A**

Q. A patient presents to the OPD with complaints of bilateral hearing loss with worsened right ear hearing loss. He is diagnosed with otosclerosis and recommended a stapedectomy as a final report to the problem. What would be the findings of tuning fork tests on this patient?

INI CET May 2023

A. Rinne's test: positive; Weber's test: lateralised to left ear
B. Rinne's test: positive; Weber's test: lateralised to right ear
C. Rinne's test: negative; Weber's test: lateralised to left ear
D. Rinne's test: negative; Weber's test: lateralised to right ear

Answer: **D**

Q. Carhart's notch is seen in which of the following conditions?

INICET Nov 2024

A. Otosclerosis
B. Ossicular discontinuity
C. Oto syphilis
D. Serous otitis media

Answer: **A**

YOURWISH

8. INNER EAR DISEASES

MENIERE'S DISEASE

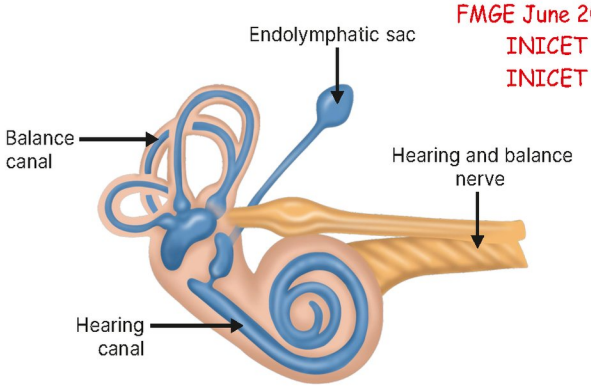
00:00:48

FMGE June 2022, 2024
INICET May 2023
INICET May 2025

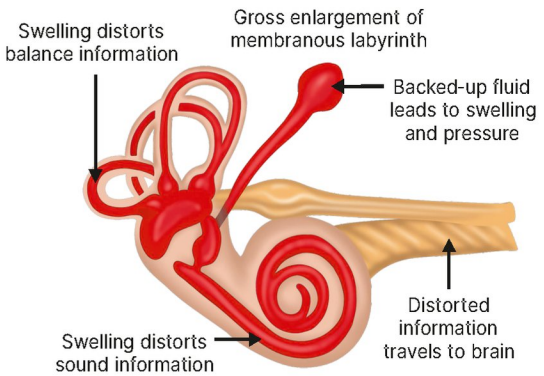
- Disorder characterized by:
  - Vertigo
  - Sensorineural hearing loss
  - Tinnitus and aural fullness

- Increased endolymphatic pressure
  - Due to ↑ production or ↓ reabsorption of endolymph at stria vascularis

Overproduction or malabsorption of endolymph
↓
Endolymphatic hypertension
↓
Gross enlargement of the membranous labyrinth
↓
Endolymphatic hydrops



Healthy Inner Ear



Meniere's Disease

- Other predisposing factors:
  - Allergy
  - Sodium and water retention
  - Hypoadrenalism and hypopituitarism
  - Hypothyroidism

Cardinal Features

Vertigo	• Minutes to hours • Severe sudden vertigo accompanied by nystagmus, nausea, vomiting, and vagal disturbances like diarrhea, cold sweat, pallor, and bradycardia
Hearing loss	• Sensorineural hearing loss • Low-frequency hearing loss • Distortion of sound (Diplacusis) • Intolerance to loud sounds due to recruitment → Tullio phenomenon • Typically fluctuating
Tinnitus	• Low-pitched roaring type
Sense of fullness of ear	• Aural fullness

Examination And Investigations

- Otoscopy: Normal
- TFT – in favour of Sensorineural hearing loss
- Nystagmus is seen only during attack.
  - The fast component is towards the diseased ear
- Pure tone audiometry: Upsloping-low frequency SNHL
- Recruitment is positive
- Glycerol test:
  - Dehydrating agent, when given orally, reduces endolymphatic pressure and improves hearing
- Caloric tests: Canal paresis
- The diagnostic test for Meniere's is: Electrocochleography
  - SP/AP ratio > 30%

Variants

- Cochlear: Present only with cochlear symptoms and no vertigo
- Vestibular hydrops: Only vertigo with no cochlear symptoms
- Lermoyez syndrome: Tinnitus f/b hearing loss and vertigo
- Tumarkin's otolith crisis: Sudden drop attack without loss of consciousness
- Secondary Meniere's:
  - Congenital and acquired syphilis
  - Paget's disease
  - Post stapedectomy

Treatment

- Conservative:
  - Low salt diet (<1.5-2g / day)
  - Stop smoking, as nicotine causes vasospasm
  - Avoid excessive intake of water
  - Correct hormonal imbalances like hypothyroidism, hypopituitarism
  - Elimination of allergens (in 50% allergens are found)
- Medical: Vestibular sedatives
- Interventional:
  - 1st line → Intratympanic steroids
  - Next → Intratympanic gentamicin
- Surgical (if medical management fails):
  - Labyrinthectomy

Surgical Management

Conservative procedures (Preserve hearing)	Destructive procedure (Destroy hearing)
• Decompression of endolymphatic sac. • Endolymphatic shunt • Sacculotomy (Ficks procedure) • Cody tack procedure • Cochleosacculostomy (cochlear duct is punctured)	• Labyrinthectomy • Intermittent low-pressure pulse therapy with Meniett device

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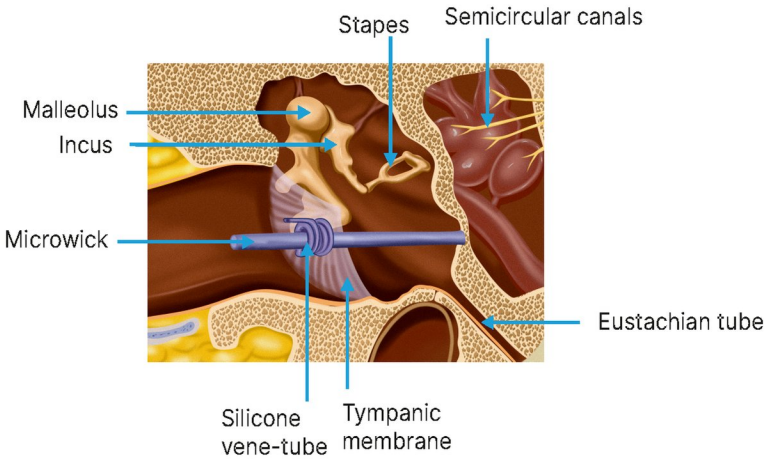
Management when medical therapy fails (intractable vertigo)

If useful hearing is still present:

- Intratympanic (transtympanic) steroid injections.
- Chemical labyrinthectomy using gentamicin (a vestibulotoxic agent), administered through the round window with a micro-wick.
- Vestibular nerve section.
- Endolymphatic sac decompression surgically guided using Donaldson's line as a landmark (the endolymphatic sac is located inferior to this line).

If there is no serviceable hearing remaining:

- Surgical labyrinthectomy



LABYRINTHITIS

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- Inflammation of the entire labyrinth
- Vertigo lasts for days to weeks

Feature	Circumscribed	Diffuse serous	Diffuse suppurative
Pathology	• Thinning or erosion of bony capsule of labyrinth • Labyrinth is exposed to external stimuli	• Diffuse inflammation of labyrinth without pus • Labyrinth is inflamed and irritable	• Pus or infection of labyrinth • Lost vestibular and cochlear functions
Reversibility	-	Reversible	Irreversible
Fistula test	Positive (communication between middle and inner ear)	Negative	Negative
Nystagmus	Positive pressure: nystagmus to same side. Negative pressure nystagmus to opposite side	Nystagmus to same side	Nystagmus to opposite side
	Transient vertigo on applying pressure to tragus	Vestibular and cochlear symptoms	Toxic + severe symptoms of serous labyrinthitis

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Differential Diagnosis Of A Patient With Vertigo And Snhl/Mixed Hearing Loss

Feature	Labyrinthitis	Perilymphatic Fistula	Meniere's Disease (Endolymphatic Hydrops)
Vertigo	Persistent vertigo lasting for several days	Triggered by coughing, straining, or loud sounds (Positive Tullio phenomenon / Hennebert sign)	Recurrent episodes lasting 20 minutes to 24 hours
Hearing loss	Sensorineural or mixed	Sensorineural or mixed; may be sudden, persistent, or fluctuating	Sensorineural hearing loss, tinnitus, and a sensation of ear fullness.
Associated features	Often preceded by or associated with a history of acute or chronic otitis media or meningitis.	• Commonly follows trauma such as stapedotomy, middle ear surgery, barotrauma, or may be associated with cholesteatoma • Ear fullness may be present	• Fluctuating hearing levels • Diplacusis (different pitch perception in each ear) • Intolerance to loud sounds, sudden drop attacks (Tumarkin crises) • Tullio phenomenon and Hennebert sign may be observed. • Sensorineural hearing loss, tinnitus, and a sensation of ear fullness. • Vertigo may be accompanied by autonomic (vagal) symptoms and sometimes an aura.

BPPV (BENIGN PAROXYSMAL POSITIONAL VERTIGO)

00:10:14

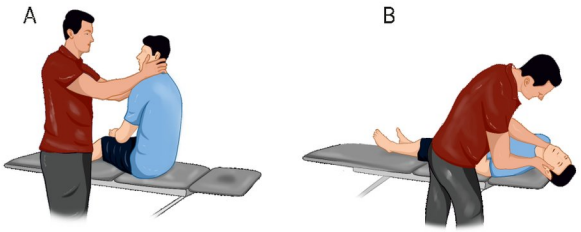
- Transient positional vertigo
- Due to the displacement of otoliths (canaliths)
  - From utricle → semicircular canal
- Cause: Trauma or sudden movement
- MC canal → Posterior semicircular canal
- Mechanism:
  - Otoliths stimulate canal receptors → sudden vertigo

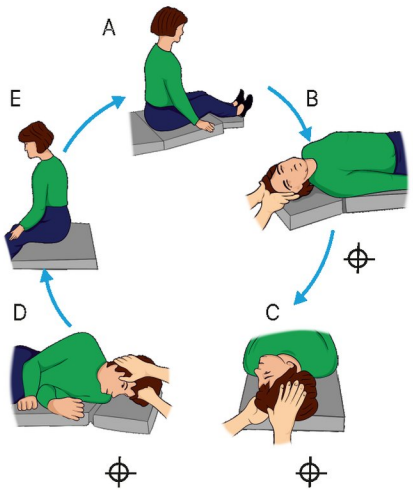
Clinical Features

- Symptoms:
  - Sudden vertigo lasting for a few seconds to minutes
- Signs: Nystagmus
  - As the cochlea is not involved → No hearing loss or tinnitus

Diagnosis

- Dix-Hallpike test
- Nystagmus pattern:
  - Posterior canal → Upbeating + torsional
  - Lateral canal → Horizontal
  - Superior canal → Downbeating





Treatment

- Epley maneuver → Canalith repositioning maneuver
  - For posterior canal involvement

VESTIBULAR NEURITIS

00:11:45

- Inflammation of the vestibular nerve only
- Sudden vertigo
  - Single long episode of vertigo lasts for a few days to weeks
- No cochlear symptoms: Because only the vestibule is involved, and cochlea is not
- Usually due to a viral infection
  - Mostly a self-limiting disease

DIFFERENTIAL DIAGNOSIS: VERTIGO WITHOUT HEARING LOSS

00:12:15

Feature	Benign Paroxysmal Positional Vertigo	Vestibular Neuritis
Definition	Transient positional vertigo Due to the displacement of otoliths	Acute inflammation limited to the vestibular nerve.
History	• Recurrent, brief episodes of spinning sensation • Each episode lasts a few seconds • Triggered by specific head movements or changes in position	• Sudden onset of severe vertigo • Single prolonged episode lasting days to weeks • Hearing remains normal • No associated central neurological signs
Examination Findings	**Dix-Hallpike Test:** • Upbeating, torsional nystagmus • Fast phase directed toward affected/hyperactive ear • Suggests posterior semicircular canal (most common)	• Peripheral type nystagmus • Fast phase directed toward the healthy (active) side • Positive head thrust test (corrective saccades present)

	• **Supine Roll Test:** • Horizontal nystagmus • Fast phase toward the involved side • Suggests horizontal semicircular canal	
Management	**For Posterior Canal:** Epley maneuver (first-line), Semont maneuver, or Brandt-Daroff exercises. **For Horizontal Canal:** Gufoni maneuver or Barbecue (log roll) maneuver.	• Short-term use of vestibular suppressants • Vestibular rehabilitation therapy

COMPARISON TABLE (HIGH YIELD)

00:14:27

Feature	BPPV	Meniere's	Labyrinthitis	Vestibular Neuritis
Duration	Seconds–minutes	Minutes–hours	Days–weeks	Days–weeks
Special feature - Vertigo	Positional	Sudden onset A/w vagal symptoms	H/O ASOM/CSOM	Post-viral
Hearing loss	No	SNHL (low frequency, fluctuating)	SNHL (constant)	No
Tinnitus	No	Yes (roaring)	Yes	No
Cause	Otolith displacement	Endolymph ↑	Infection	Post-viral
Diagnosis	Dix-Hallpike Supine roll test	ECOG	Caloric tests/ VEMP	
Treatment	Epley maneuver Barbecue (log roll) maneuver	Intratympanic Steroids → Gentamicin → Surgery	Antibiotics Clearance of disease	Antibiotics/ Antivirals Supportive

PYQ

00:16:53

Q. What could be the potential clinical diagnosis for a 35-year-old man who visited the clinic reporting intermittent dizziness, sensation of pressure and fullness in the ear, reduced hearing, and a low-frequency roaring sound in the ears?

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A. Meniere's Disease
B. Benign Paroxysmal Positional Vertigo
C. Labyrinthitis
D. Syncope

Answer: A

Q. A 50-year-old male presented with vertigo, tinnitus, and hearing loss. The audiogram is given below. What is the most likely diagnosis based on the given image?

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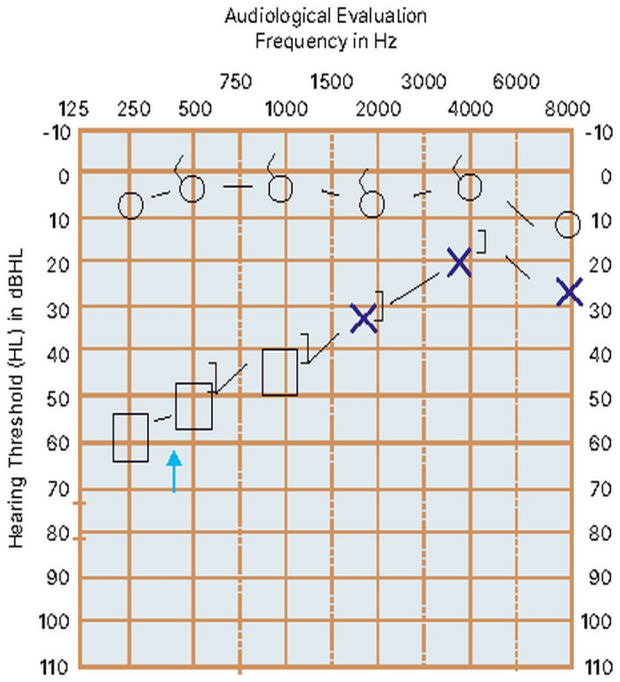
A. Bppv
B. Meniere's Disease
C. Vestibular Neuronitis
D. Presbycusis

Answer: B



Q. A patient presents to the OPD with complaints of vertigo, which is sudden in onset, sensorineural hearing loss (SNHL), and tinnitus, which lasts minutes to hours, accompanied by nausea, vomiting, and vagal symptoms. Investigations show that endolymphatic hypertension is present. What is the diagnosis of the patient?

A. Meniere's Disease
B. Presbycusis
C. Benign Paroxysmal Positional Vertigo (Bppv)
D. Acoustic neuroma

Answer: A

Q. A 45-year-old man complains of early morning vomiting that occurs after turning in bed. What is the most likely diagnosis?

A. Benign Paroxysmal Positional Vertigo (Bppv)
B. Meniere's Disease
C. Vestibular Neuritis
D. Labyrinthitis

Answer: A

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9. EXTERNAL EAR DISEASES

CONGENITAL ABNORMALITIES OF PINNA

00:00:50

Microtia

- Pinna is completely small

Macrotia

- The pinna is completely big

Anotia

- Definition: Complete absence of pinna (no pinna at all)
- Hearing rehabilitation: BAHA (same reason as microtia)

Bat Ear

- Protrusion of the pinna

Hearing Rehabilitation

- BAHA (Bone Anchored Hearing Aid)
- Cannot accommodate a regular hearing aid due to the small/absent pinna
- BAHA bypasses the external & middle ear and transfers the sound to the inner ear

Pre-Auricular Sinus

- Embryological basis:
  - The pinna develops from 6 tubercles called Hillocks of His
  - 1st hillock → Tragus
  - Remaining hillocks → Rest of pinna
  - Incomplete fusion of 1st hillock with rest → Pre-auricular sinus
- **Clinical features:**
  - A small pit or hole near the ear
  - Can get recurrently infected
  - Results in purulent discharge from the pit/hole
- **Treatment:**
  - Repeated infections = Indication for surgery
  - Otherwise: No surgery required

ACQUIRED ABNORMALITIES OF PINNA

00:02:25

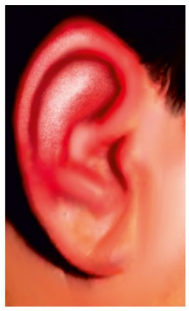
Haematoma Auris (Boxer's Ear)

- Repeated trauma to the ear (common in boxers)
- Accumulation of blood between the cartilage and the overlying perichondrium
- Clinical features
  - Swelling with pain
  - Bluish discoloration of the pinna

Treatment

- Incision and drainage
- Pressure packing (to prevent re-accumulation of blood)
- Antibiotics (mandatory)
  - Infection of haematoma → Perichondritis → Cartilage infection → Necrosis → Deformity of the pinna

Perichondritis



- Infection of the entire pinna
- Common causative organism: Pseudomonas
- Clinical features
  - The entire pinna is red, stiff, and painful
  - Hallmark sign: Sparing of the lobule
  - As the Lobule does not have cartilage and is made up of only fat

Keloid

Etiology

- Trauma (iatrogenic or accidental)
  - Iatrogenic: Surgical incision
  - Accidental: Ear piercing

Pathophysiology

- Abnormal epithelialization
- Abnormal growth of epithelium
- Hypertrophy of tissue
- Very high recurrence rate

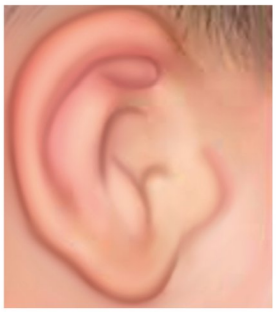
Treatment

- To prevent recurrence
- First-line therapy:
  - Intralesional hyaluronidase injections
  - Intralesional steroids
  - 8-10 injections at 4-5 day gap each
- If recurs after first-line:
  - Short course of radiotherapy
  - Short course of chemotherapy

CONGENITAL ABNORMALITIES OF THE EXTERNAL AUDITORY CANAL

00:04:45

Complete Atresia



- No canal at all
- **Rehabilitation:**
  - Cannot put in a hearing aid
  - **BAHA** (Bone Anchored Hearing Aid) required

ACQUIRED ABNORMALITIES OF THE EXTERNAL AUDITORY CANAL

00:05:00

Wax (Cerumen)

- The most common acquired cause of hearing loss in adults

Source

- Secretions from the sebaceous glands of the external auditory canal
- Specifically from the cartilaginous portion

Function

- **Acidic pH**
- Protects from microorganisms (bacteria, viruses, other microorganisms)
- Prevents anything from reaching the tympanic membrane

Causes Of Accumulation

- Excessive production
- Narrow canal → Cilia cannot beat wax completely outside

Precipitating Factor

- Water entry (during bathing/swimming)
- Wax swells up

Clinical Features

- Pain
- Hearing loss
- Ear fullness
- Tinnitus (ringing sensation in the ear)

Treatment

- Wax-solytic agents
- Manual syringing

OTITIS EXTERNA

00:06:07

Classification

Classification Basis	Types
Based on the extent	Localized / Diffuse
Based on duration	Acute / Chronic
Based on the cause	Infectious (Bacterial / Viral) / Non-infectious (Eczema, Seborrheic)

Localized Otitis Externa (Furunculosis)

- Staphylococcal infection of the hair follicle
- Cartilaginous portion of the external auditory canal
- **Clinical features:**
  - Pain is increasing with jaw movement
  - Other features:

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→ Pain on touching the ear
→ Pain on chewing movement

- **On examination**
  - Tragal sign positive (pressing tragus → patient winces in pain)
  - Purulent discharge
    → Purulent = Pathology in the external auditory canal
    → Mucoid/mucopurulent = Disease in the middle ear
- **Retroauricular groove:** Not obliterated (unlike mastoiditis, where there is deepening)
- **Treatment**
  - Oral antibiotics
  - Aural toileting
  - If recurrent furunculosis:
    → Look for immunosuppression or diabetes

Diffuse Otitis Externa

00:09:23

- Swimmer's ear
- Singapore ear
- Seen in immunocompetent patients

> Important Information
> - Pseudomonas infection in immunocompetent = Diffuse otitis externa
> - Pseudomonas infection in immunocompromised = Malignant otitis externa

- Swimmer's ear/Singapore ear/Telephone ear:
  - Exposure to swimming/water
  - Exposure to moisture
  - Excessive use of the telephone
  - Tropical countries (Singapore/Malaysia) → High humidity

- **Pathophysiology:**
  - Normal: Wax pH = **Acidic** (protective)
  - Exposure to humidity/water/moisture → pH becomes **Alkaline**
  - Alkaline pH → Favors bacterial growth
  - Result: Infection in the canal
- **Clinical features:**
  - Pain
  - Purulent discharge
  - Hearing loss
  - Pain increases on jaw movement (like localized otitis externa)
- **Examination findings**
  - The entire external auditory canal is inflamed and edematous (vs. localized swelling in furunculosis)
  - Tragal sign present
- **Treatment:**
  - Anti-pseudomonal antibiotics
  - Aural toileting

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Malignant Otitis Externa

00:11:30

- Not a malignancy - called "malignant" because its very aggressive
- Elderly patient
- Diabetic patient
- Immunosuppressed patient
- Cause: Pseudomonas
- Clinical features:
  - Excruciating pain - unable to sleep at night
  - Purulent discharge
  - Hearing loss (HOH)
  - Sometimes fever
- Examination
  - Hallmark sign- Granulation tissue at the bony-cartilaginous junction of the external auditory canal
  - Cranial nerve involvement:
  - Initially: 7^{th} nerve (Facial nerve palsy)
    → Reason: Pseudomonas is a bone-eating organism
    → Causes osteoclastic activity
    → Erodes bone
  - Later: Base of skull involvement → 9^{th}, 10^{th}, 11^{th} nerve palsy

Investigations for malignant otitis externa:

- Culture-To confirm Pseudomonas
- Technetium-99 scan
  - The earliest diagnostic test
  - Radio-isotopes picked up by osteoclasts
  - Positive even at the beginning/very early stage
  - Positive even when bony erosion is not yet visible on the CT scan
  - Most important for early diagnosis
- CT scan
  - Identifies bony erosion
  - Used to identify the extent of the disease
  - But not as early as Technetium-99
- MRI- To identify the intracranial extension of the disease
- Gallium-67 scan
  - For monitoring prognosis
  - To decide whether to continue or discontinue therapy
- Mechanism:
  - Gallium is taken up by neutrophils
  - Neutrophils = Cells of acute inflammation (NOT chronic)
  - Initially (active disease): Scan positive
  - After treatment: Neutrophils disappear → Scan becomes negative → Resolution
  - If scan still positive: Neutrophils still present → Continue therapy

Management Of Malignant Otitis Externa

- Treatment:

- Anti-pseudomonal antibiotics
- Topical therapy
- Symptomatic therapy

- Follow-up:
  - Repeated cultures
  - Repeated scans to identify if the disease is:
    → Localized and regressing, OR
    → Still extensive/progressing

OTOMYCOSIS

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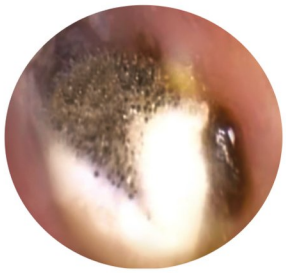
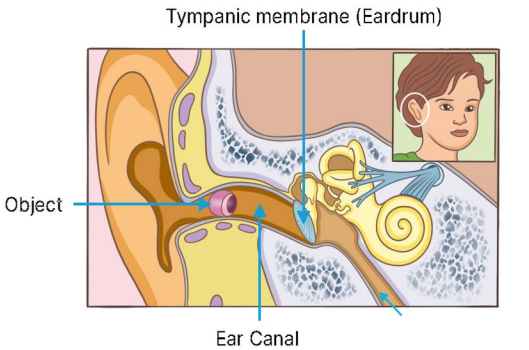


- Common causative organisms:
  - Aspergillus niger (Most common)
  - Candida
  - Aspergillus fumigatus
- Characteristic appearance: "Wet newspaper appearance"
  - White discharge
  - Black spores overlying
  - Candida cotton ball appearance
- Clinical features:
  - Most important complaint: Itching
  - Discharge
  - Pain
  - May have secondary bacterial infection
- Treatment:
  - Topical antifungal
  - Aural toileting (clean the ear, make it dry)
  - patient education on the importance of keeping the ear dry

FOREIGN BODY IN EXTERNAL AUDITORY CANAL

00:17:03

- Common in Toddlers and young children
- Clinical presentation: U/L foul-smelling discharge (ear or nose) in children → Signifies foreign body unless told otherwise



Methods Of Removal

- Inorganic bead/pellet: Remove with the probe
- Graspable non-living object (paper, rubber): Use crocodile forceps
- Living insect:
  - First asphyxiate (to prevent movement)
  - Do not use oil (not recommended)

- Use 4% Xylocaine drops- Makes the organism immobile & asphyxiate (cannot get oxygen), then remove the insect

- Organic foreign body OR Button battery:
  - NEVER DO SYRINGING
  - Reasons:
    → Organic FB: Absorbs water → Swells up → Further occlusion
  - Button battery: Requires urgent removal with a probe

Syringing Technique

- Instrument: Simpson's aural syringe
- Water temperature: 37°C (exactly)
  - Not above or below
  - Can cause caloric reaction → Syncope
- Direction of ear pull – Posterosuperior
- Direction of water jet – Posterosuperior
  - Straight direction → Tympanic membrane perforation
- Contraindications to syringing:
  - Hygroscopic foreign body
  - Organic foreign body
  - Button battery
  - Tympanic membrane perforation
- Complications:
  - Vertigo (if water is not at body temperature, due to caloric stimulation)
  - Rupture of TM (if done too forcefully)
  - Reactivation of quiescent otitis media.

KERATOSIS OBTURANS

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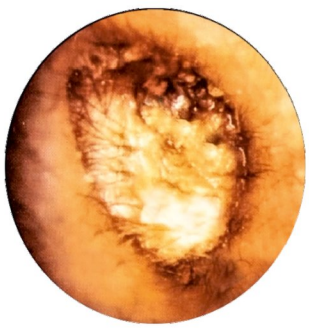

- Mass of keratin accumulating in the external auditory canal
- **Pathophysiology:**
  - Normally, cilia push dead epithelium outside
  - In keratosis obturans: Ciliary motility disorder
  - Dead epithelium cannot be pushed out
  - Layers of keratin/epithelium are organized one over another
  - "Onion skin/lamellar arrangement"
  - Effects:
    → Causes widening (NOT erosion)
    → Due to widening and pressure → 7th nerve palsy possible
- Clinical features:
  - **Severe otalgia** (extremely painful)
  - Ballooning of the canal
- Examination
  - White mass with wax in the deep meatus, ulceration, granulations, and widening of the EAC. Facial nerve palsy.
  - History of chronic sinusitis and bronchiectasis.
- Diagnosis: Clinical
- Treatment:

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- Soften with wax-solytic agents
- Removal requires sedation or short GA (due to severe pain)
- Removal by instrumentation

HERPES OTICUS (RAMSAY HUNT SYNDROME)

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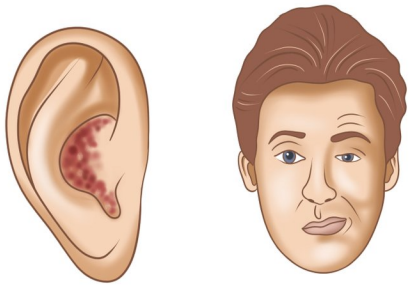
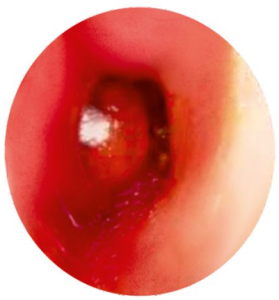


- Causative organism: Herpes zoster virus (Varicella-zoster virus)
- Pathophysiology:
  - Virus remains latent in the geniculate ganglion
  - Reactivation occurs due to immunosuppression (Not low immunity)
- Clinical features:
  - Pain in the area of distribution of the facial nerve
  - Vesicles in the area of distribution of the facial nerve
  - LMN type of facial nerve palsy
- Treatment:
  - Antivirals
  - Steroids
- Prognosis:
  - Poorer than Bell's palsy
  - Not as good a recovery as in facial palsy

DISEASES OF THE TYMPANIC MEMBRANE

00:21:40

Myringitis Bullosa Hemorrhagica



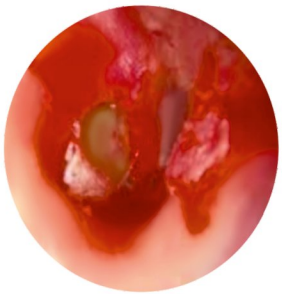
- Association: Often associated with acute otitis media
- Causative organisms:
  - Streptococcus pneumoniae
  - Mycoplasma
- Clinical features:
  - Pain
  - Fever
  - Hearing loss
  - Vesicles/blebs on the tympanic membrane
  - Sometimes, blood-stained discharge (if the vesicle ruptures)
- Investigations: NOT required
- Examination is diagnostic
- Treatment:
  - Antibiotics against Streptococcus and Mycoplasma
  - Symptomatic therapy

Granular Myringitis

- **Definition:** Chronic infection of the tympanic membrane
  - Tympanic membrane is replaced by **granulation tissue**
- **Causes:**
  - Bacterial infection
  - Fungal infection
  - Trauma

  - Post-surgery
- **Clinical features:**
  - Blood-stained ear discharge
  - Pain
  - Itching in the ear
- **Examination:** Granulations over the tympanic membrane
- **Treatment:**
  - Topical antibiotics + Systemic antibiotics
  - Cauterization (very important - to prevent further granulations)

Traumatic Perforation



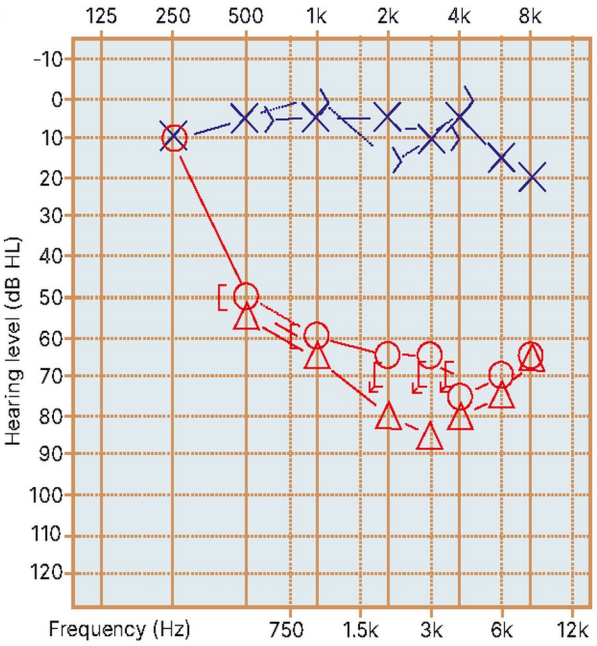
- Etiology: Trauma causing a hole/perforation in the tympanic membrane
- Clinical features:
  - Pain
  - Hearing loss
  - Blood-stained discharge
- Investigations: NOT required
- On examination: Perforation visible with blood and retracted parts
- Treatment:
  - Wait and watch
  - Will close by itself in 6-12 weeks
- If it does NOT close:
  - Surgery: Myringoplasty

10. HEARING LOSS

SUDDEN SENSORINEURAL HEARING LOSS (SSHL)

INICET May 2024, FMGE Jan 2025

- Definition: >30 dB hearing loss for 3 consecutive frequencies within 3 days (72 hours)
- Causes: Viral, autoimmune, post-infection tumors like acoustic neuroma, exposure to loud sounds
  - Conductive: Wax in the ear
  - Sensorineural: Cochlear or Retrocochlear
- Diagnosis
  - Audiometry→ MRI with contrast (used to rule out intracranial pathologies compressing the nerve).
  - The audiogram shows a right ear drop in hearing levels to 50, 60, 70 dB at 500, 1K, 2K Hz, respectively
- Treatment: The earlier the diagnosis and treatment, the better the prognosis
  - Steroids are the main Rx
  - Intratympanic Steroids and Systemic Steroids



Keys to Symbols	Right	Unspecified	Left
AC unmasked	O		X
AC masked	●		⧗
BC unmasked	<		>
BC masked	[	△	]
AC aided	H	□	V
AC sound field			

Inner Ear Condition Causing Isolated Hearing Loss
(External ear and middle ear including TM are normal → Pure SNHL)

Clinical Presentation

- Patient presents with complaints of sudden hearing loss developing over a 3-day period.

Diagnostic Evaluation

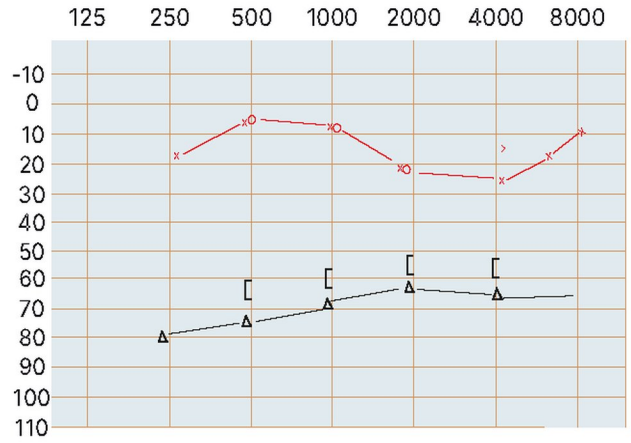
- Rinne test: May be positive or negative (False negative can occur in severe SNHL)
- Diagnostic dilemma: Severe SNHL may be mistaken for CHL on tuning fork tests due to a negative Rinne.
- Pure Tone Audiometry (PTA):
- >30 dB SNHL in three contiguous frequencies

Diagnosis

- Sudden Sensorineural Hearing Loss (SNHL) – an otologic emergency
- Definition:
- Hearing loss >30 dB affecting three consecutive frequencies within 3 days

Management

- Medical treatment:
  Oral steroids, intratympanic steroids, carbogen therapy, hyperbaric oxygen
- **Rehabilitation:**
  If unilateral deafness → BAHA or CROS hearing aids
- **Imaging:**
  MRI to exclude acoustic neuroma



Important Information

- Cross hearing: A hearing aid that transmits the sound from the ear with severe hearing loss to the opposite normal ear and the brain
- >90 dB is a loud sound
- >150 dB causes acoustic trauma and permanent deafness

Conditions of the Inner Ear Causing Only Hearing Loss

Noise-Induced Hearing Loss (NIHL)

- History of hearing loss occurring after sudden or prolonged exposure to loud noise
- May result in a temporary or permanent threshold shift

Ototoxicity

- History of exposure to ototoxic agents, such as: aminoglycosides, antimalarials, analgesics, cytotoxic drugs, diuretics, vancomycin, erythromycin, or azithromycin

Presbycusis

- Occurs in individuals older than 50 years
- Hearing loss is slowly progressive, bilateral, and symmetrical
- Most common cause of SNHL in adults

Diagnostic Tests

- Pure Tone Audiometry (PTA): Down-sloping curve indicating high-frequency hearing loss
- Acoustic dip at 4 kHz characteristic of NIHL
- High-frequency audiometry: Demonstrates high-frequency hearing loss
- Otoacoustic emissions (OAE): Absent in NIHL and ototoxicity due to early involvement of outer hair cells

Management

- Hearing aids
- Cochlear implant

Frequently Asked PYQs

- Safe noise exposure limit: 85 dB for 8 hours/day, 5 days/week
- Prevention of NIHL: Ear plugs (30 dB protection) and ear muffs (40 dB protection)
- Aminoglycosides and cytotoxic drugs cause irreversible hearing loss
- Cochleotoxic aminoglycosides (Mnemonic: KAN) — Kanamycin, Amikacin, Neomycin (most toxic)
- Streptomycin, Gentamicin, Tobramycin are predominantly vestibulotoxic
- Loop diuretics block the Na^+-K^+-$2Cl^-$ channel of the stria vascularis
- Most common type of presbycusis: Neural
- Other types include sensory, cochlear conductive, and strial

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CHILDHOOD HEARING LOSS

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Etiology of Childhood Hearing Impairment

Congenital Causes of Hearing Loss (Prenatal) — 2/3

Genetic Causes (50%)	Non-Genetic Causes (50%)
Non-syndromic: 70%	Maternal infections: CMV (most common), rubella, syphilis, toxoplasmosis
Syndromic: 30%	• Drug exposure and maternal factors: • Use of ototoxic drugs, substance abuse, and endocrine disorders during pregnancy

- The above factors affect **labyrinth development during the first trimester**.

Acquired Causes (Perinatal and Postnatal) — 1/3

- Infections:
  Meningitis (most common), measles, mumps, varicella, herpes, toxoplasmosis, chronic otitis media
- Hypoxia:
  ARDS, prolonged difficult labor, prematurity
- Hyperbilirubinemia
- Ototoxic drugs
- Noise trauma / head trauma
- NICU admission for more than 5 days:
  Associated with a high incidence of Auditory Neuropathy Spectrum Disorder

AUDITORY NEUROPATHIC SPECTRUM DISORDER

- In auditory neuropathy spectrum disorder, patients will hear sounds but have poor speech discrimination

Auditory Neuropathy Spectrum Disorder (ANSD)
A higher incidence of ANSD is observed in **neonates admitted to the ICU**.

Pathophysiology of ANSD – Important Concept

- Dysfunction occurs at the level of the cochlear inner hair cells (IHC), resulting in inadequate transmission of signals to the auditory nerve.
- Demyelination of the auditory pathway nerves is seen.
- There is loss of axons.
- This leads to dyssynchrony.
  Sound continues to travel, reach, and be perceived by the auditory cortex; however, speech (complex sounds) cannot be identified due to loss of synchronization.

Presentation of a Child with ANSD

- The child is able to hear sounds and responds to sound.
- However, the child does not understand complex sounds, such as speech, which require synchronization.
- Hence, the condition is usually detected when the child is slightly older.

Auditory Neuropathy Spectrum Disorder (ANSD)

Test Results Pointing Toward ANSD

- Pure Tone Audiometry (PTA): Normal or mild to moderate SNHL, as the child is able to perceive sound stimuli.

Management

- Hearing aids
- Cochlear implant

YOURWISH

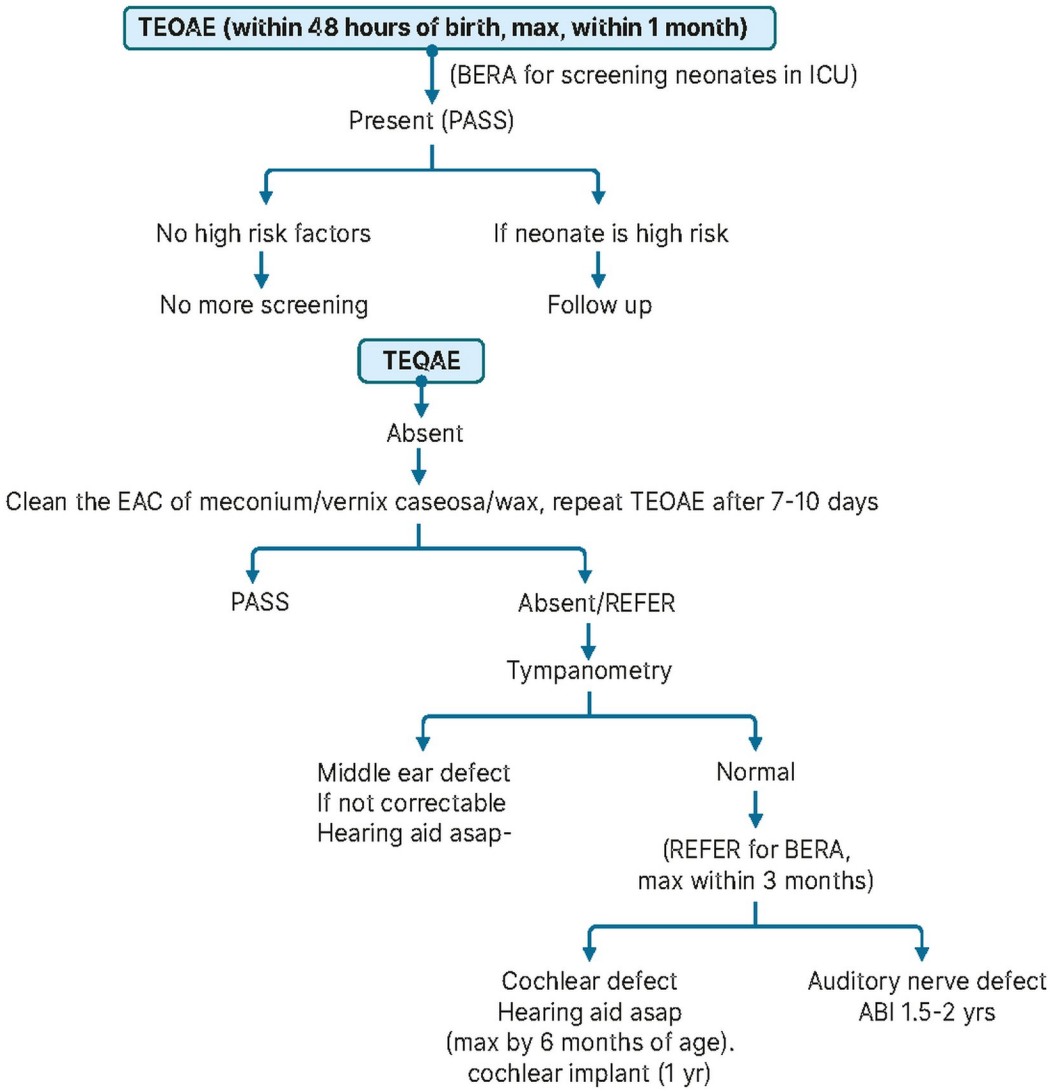
YOURWISH

- Speech audiometry / speech recognition / speech discrimination: Speech intelligibility is disproportionately poorer compared to the degree of hearing loss.
- Otoacoustic emissions (OAE): Present, as outer hair cells are normal; therefore, BERA is used for hearing screening in NICU neonates, since OAE may be normal (PASS) and ANSD can be missed.
- **BERA:** Abnormal.
- **MLR (middle latency response: 12–50 ms)** and **LLR / cortical responses (50–500 ms)** assessed by **CERA** may be abnormal or absent

NEONATAL HEARING SCREENING

- Every child at birth should undergo an auto acoustic emission test
- If OAE is normal, then the patient passes, and no higher tests are required

Protocol for Neonatal Hearing Screening



1-3-6 rule-screening test within 1 month of birth, referral test within 3 months, rehabilitation within 6 months.

BEHAVIOURAL OBSERVATION AUDIOMETRY

- Done in those children in whom tuning fork tests cannot be done

Behavioural Observation Audiometry (BOA)

BOA is a test used to assess a child's hearing by observing changes in the child's behavior or response following the presentation of a sound stimulus.

When to Perform BOA

BOA is performed in children older than 6 months who present with complaints of hearing loss.

↓

This is followed by detailed history taking and clinical examination.

↓

If findings are normal up to the tympanic membrane, BOA is carried out.

↓

If BOA findings are abnormal or if the child is unresponsive, the child is subjected to electrophysiological tests, starting with OAE, followed by BERA.

Behavioural Observation Audiometry (BOA)

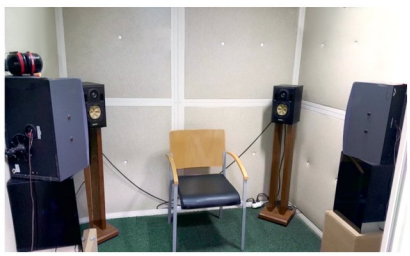
Important Tests to Remember in BOA

Age:	Developmental milestone:	BOA tests:
5-24 Months	The child begins to localize sounds and turns the head toward the sound source.	· Free field audiometry · Visual reinforcement audiometry
2-5 Years	The child develops some vocabulary and is able to follow simple instructions.	Play audiometry

Note : When the child is older than 5 years, preferably above 7 years, Pure Tone Audiometry (PTA) can be performed.

PLAY AUDIOMETRY

- The child is made to sit in the chair
- Music is played from the speakers surrounding
- If the child localises the sound, hearing is intact



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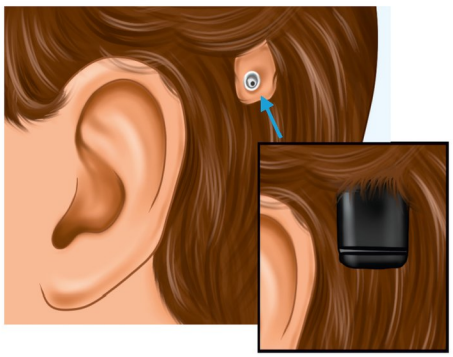
REHABILITATION

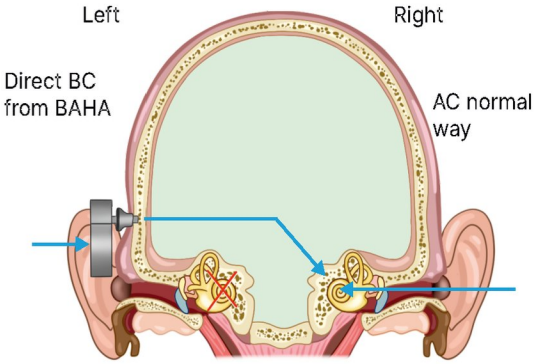
Hearing loss	Rehabilitation
Moderate to moderately severe SNHL/CHL	Hearing aid
External auditory canal abnormality / chronic discharging ear in an old patient	BAHA
Unilateral severe to profound SNHL	BAHA
Bilateral severe to profound SNHL	Cochlear implant
Otosclerosis	Surgery

BAHA- BONE ANCHORED HEARING AID

- A titanium screw is integrated into the squamous temporal bone
- Once it is integrated with the bone(usually in 6 months), the external component is put to fit over the screw
- The processor sends sound to the opposite ear and also to the same ear
- Hence, it is used in U/L sensorineural hearing loss

NEET PG 2018





COCHLEAR IMPLANTS

Cochlear Implant

Inventor

- Developed by **William F. House**, who is regarded as the **Father of Neurotology**.

Indications

- **Severe to profound hearing loss in both ears** with **no significant benefit from hearing-aid use**.
- Implantation can be considered **from 1 year of age onwards**.

Contraindications

- **Non-formation of the cochlea** (Michel aplasia)
- **Absence of the cochlear nerve**
- Presence of **central auditory pathway lesions**

Components of the Implant

Internal Components

- Receiver–stimulator unit
- Electrode array

External Components

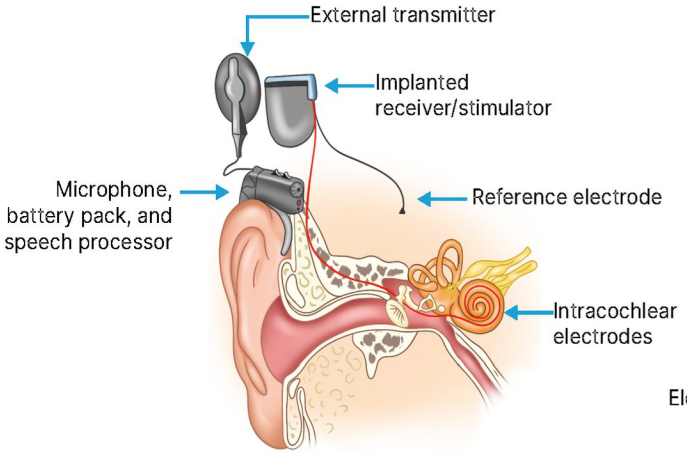
- Microphone
- Speech processor
- Transmitting coil

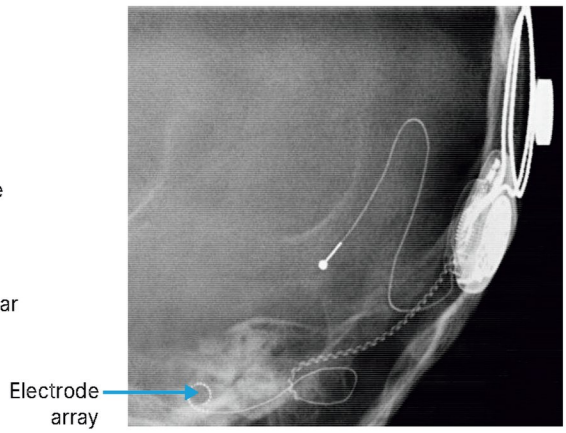
Commonly Asked PYQs

- The procedure may be performed **starting at 1 year of age**.
- Entry into the **middle ear** is achieved via the **facial recess**.
- Entry into the **inner ear** occurs through the **round window**.
- The electrode is positioned in the **scala tympani**.
- It functions as a substitute for the **organ of Corti**.
- Neural activation occurs at the level of the **cochlear nerve**.

- Has an external and internal component
- The external component has a microphone and the speech processor
- It transfers the sound to the internal component (Receiver/speech stimulator)
- The reference electrode is in the squamous temporal region
- The internal electrode is in the scala tympani of the cochlea

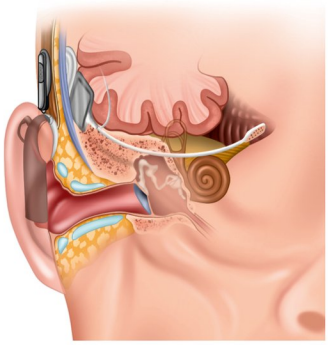
YOURWISH





AUDITORY BRAINSTEM IMPLANT

- Used in patients with B/L acoustic neuroma and undergone surgery/absent or dysfunctional 8th nerve



DEFINITION OF DEAF

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- Calculating Hearing Disability
- Calculate Monaural (One Ear) Loss:
  - Perform this for the left and right ears individually:
  - Determine Pure Tone Average (PTA): Add the hearing thresholds at 500, 1000, 2000, and 3000 Hz and divide by 4
  - Apply the "Low Fence": Subtract 25 dB from the average (as 25 dB is considered the limit of normal hearing)
  - Multiply by 1.5: Every decibel above the 25 dB baseline equals 1.5%
  - Formula: (PTA-25) x 1.5= % Monoaural impairment
- Calculate Binaural (Both Ears) Loss
  - To find the total disability percentage, the "better ear" is weighted more heavily because it contributes more to daily functioning.
  - Weighted Average:
  - Formula: [(% Better Ear x 5) + (% Worse Ear)] ÷ 6 = % Binaural Hearing Disability

PYQs

00:15:45

Q. A patient presents to the OPD with hearing loss. On examination: Rinne test - Positive in the left ear and negative in the right ear.Weber's test - Lateralization towards the left.Bone conduction test - Reduced in the right ear and similar in the left ear

FMGE Jan 2025

Interpret the findings:

A. Sensorineural hearing loss in the left ear

B. Sensorineural hearing loss in the right ear

C. Conductive hearing loss in the left ear

D. Conductive hearing loss in the right ear

Answer: **A**

- Rinne test: Right ear negative → suggests Conductive hearing loss (CHL)
- Weber test:
  - Sound goes to the CHL ear or better ear in SNHL
  - Here → Lateralizes to left→ Either left CHL OR right SNHL
- Bone conduction:
  - Reduced in right ear → Confirms SNHL in right ear

Q. Which of the following interventions should not be done in an 8-year-old boy with bilateral sensorineural hearing loss?

INICET May 2024

A. Preferential sitting in the classroom
B. Adenoidectomy with grommet insertion
C. Hearing aid
D. Cochlear implant

Answer: **B**

- Adenoidectomy with grommet insertion is done in conductive hearing loss